

Question	Answer	Marks
4(a)(i)	(a wave that) transfers / propagates energy	B1
4(a)(ii)	distance = $n\lambda$	B1
	time $t = n / f$	B1
4(a)(iii)	(v = distance / time) (v) = $n\lambda / (n / f)$ so (v) = $f\lambda$	B1
4(b)(i)	$f = 3 \times 10^8 / 4 \times 10^{-2}$ $= 7.5 \times 10^9$ (Hz) $= 7.5 \times 10^9 / 10^9$ (GHz)	C1
	= 7.5 GHz	A1
4(b)(ii)	(difference in path lengths, $XQ - YQ$) = $\left[72^2 + 30^2\right]^{\frac{1}{2}} - 72 = 6$ (cm)	A1
	Or calculate angle Q using tan and then cos to obtain XQ of 78 $\tan \theta = 30 / 72$ so $\theta = 22.6$ $XQ = 72 / \cos 22.6 = 78$ (cm) path difference $78 - 72 = 6$ (cm)	(A1)
	Or calculate angle X using tan and then cos to obtain XQ of 78 $\tan \theta = 72 / 30$ so $\theta = 67.4$ $XQ = 72 / \sin 67.4 = 78$ (cm) path difference $78 - 72 = 6$ (cm)	(A1)
4(b)(iii)	path difference (= 6 cm / 4 cm) $\lambda = 1.5\lambda$	M1
	(so) phase difference (at Q) = 540° or 180°	M1
	(so) intensity is minimum	A1